

**Total: 40 points****1. Expected final answer:**

$$2\mathbf{i} - \mathbf{j} + 3\mathbf{k}.$$

**Verification:** The coefficients of  $\mathbf{i}$ ,  $\mathbf{j}$ , and  $\mathbf{k}$  are 2,  $-1$ , and 3, respectively.**Equivalent forms:**  $2\mathbf{i} + (-1)\mathbf{j} + 3\mathbf{k}$  and algebraically identical standard-basis notation receive full credit.**Grading (10 points):** 3 points for each correct coefficient and sign; 1 point for valid standard-basis notation.**2. Expected final answer:**

$$-1.$$

**Verification:**

$$\vec{u} \cdot \vec{v} = 1(3) + 2(-1) + (-1)(2) = 3 - 2 - 2 = -1.$$

**Equivalent forms:** Any numerical expression equal to  $-1$  receives full credit.**Grading (10 points):** 5 points for the three component products, 3 points for combining them correctly, and 2 points for the final value.**3. Expected final answer:**

$$\vec{r}(t) = (1, -1, 2) + t(2, 2, -1), \quad t \in \mathbb{R}.$$

**Verification:**  $Q - P = (2, 2, -1)$ ; the formula gives  $P$  at  $t = 0$  and  $Q$  at  $t = 1$ .**Equivalent forms:** Any affine parametrization  $\vec{r}(t) = t\vec{v} + \tilde{P}$  of the same line receives full credit:  $\tilde{P}$  must lie on the displayed line and  $\vec{v}$  must be a nonzero scalar multiple of  $(2, 2, -1)$ . Equivalent component equations are also valid.**Grading (10 points):** 4 points for a valid direction vector, 4 points for a point on the line, and 2 points for a complete parametric equation.**4. (a) Expected final answer:**

$$\overrightarrow{PQ} \times \overrightarrow{PR} = (1, -1, 1).$$

**Verification:**

$$\overrightarrow{PQ} = (1, 1, 0), \quad \overrightarrow{PR} = (0, 1, 1),$$

$$\overrightarrow{PQ} \times \overrightarrow{PR} = (1(1) - 0(1), 0(0) - 1(1), 1(1) - 1(0)) = (1, -1, 1).$$

**Equivalent forms:** Ordered-triple and standard-basis notation are both valid. Because the order of the cross product is specified, a different nonzero scalar multiple is not the requested cross product.**Grading (5 points):** 2 points for the two edge vectors and 3 points for the correct cross product.**(b) Expected final answer:**

$$x - y + z = 2.$$

**Verification:** With  $\vec{n} = (1, -1, 1)$  and  $P$ ,

$$\vec{n} \cdot (x - 1, y, z - 1) = 0 \iff x - y + z = 2.$$

Substitution confirms that  $P$ ,  $Q$ , and  $R$  each satisfy the equation.**Equivalent forms:** Any algebraically equivalent equation, including any nonzero scalar multiple of  $x - y + z - 2 = 0$ , receives full credit.**Grading (5 points):** 2 points for using a valid normal, 1 point for substituting a point correctly, and 2 points for a correct plane equation.